

Quantum Protocols for Topological Codes with ZX Calculus

Salvador Márquez Grima

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Introduction

- The extreme fragility of quantum information requires the use of methods to detect and correct errors.
- Main difficulties: no-cloning theorem; measurement destroys information; and continuous errors.
- Solutions: Redundant encoding of information; multi-qubit parity operators; and discretization of errors

Quantum Error Correcting Codes

- We are encoding k logical qubits into n physical qubits, with $n > k$, by imposing some constraints on the physical qubits and thus defining a 2^k -dimensional subspace of the total Hilbert space, called the code subspace. [7]
- Errors push the state out of the code subspace. If each error corresponds to different undeformed and orthogonal subspaces, we can detect and correct them.

Stabilizer Codes

Definition

We define the n -dimensional Pauli group $\mathcal{P}_n$ as the set of all n -fold tensor products of Pauli matrices and the identity, with overall phases $\pm 1, \pm i$. A stabilizer is some abelian subgroup $\mathcal{S} \subseteq \mathcal{P}_n$ that does not contain $-I$

The stabilizer can determine a code subspace, called the stabilizer code $C(\mathcal{S})$, as the set of all states that are fixed by the action of all elements of $\mathcal{S}$:

$$C(\mathcal{S}) \equiv \{|\psi\rangle \in \mathcal{H} : g|\psi\rangle = |\psi\rangle; \forall g \in \mathcal{S}\}$$

Proposition

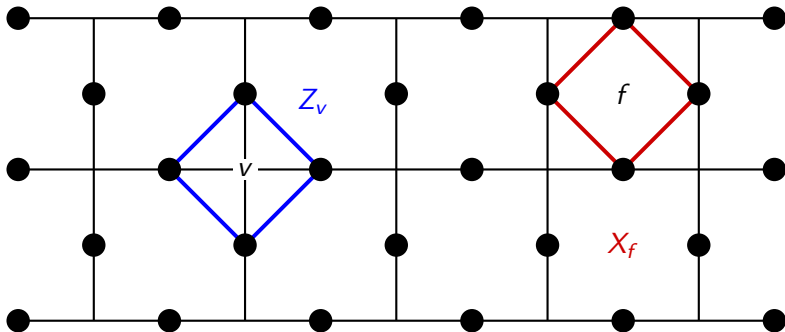
If $\mathcal{S}$ is an Abelian subgroup of $\mathcal{P}_n$ that does not contain $-I$, then the code space $C(\mathcal{S})$ stabilized by $\mathcal{S}$ has dimension 2^{n-k} , where k is the number of independent generators of $\mathcal{S}$. [4]

Topological Codes

Consider a two-dimensional lattice embedded in a closed surface, with physical qubits attached to each edge. We can define a chain complex of vector spaces over $\mathbb{Z}_2$ by constructing the standard sets of i -chains C_i for $i = 0, 1, 2$, where the basis elements are the vertices, edges, and faces of the lattice. The boundary operators $\partial_i : C_i \rightarrow C_{i-1}$ are defined as usual. [2]

We can identify elements of the computational basis with 1-chains on the lattice $|c\rangle = \bigotimes_i |c_i\rangle$, where the field of coefficients is $\mathbb{Z}_2$. Homomorphisms from C_1 to $\mathcal{P}_n$ are defined as follows:

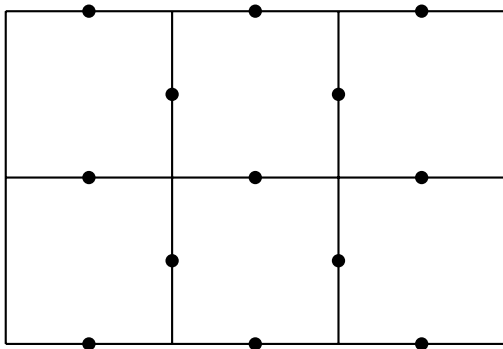
$$X_c = \bigotimes_i X_i^{c_i} \quad \text{and} \quad Z_c = \bigotimes_i Z_i^{c_i}.$$



The stabilizer generators are:

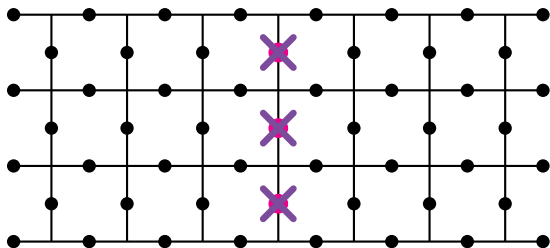
$$X_f := \prod_{e \in \partial_2 f} X_e \quad \text{and} \quad Z_v := \prod_{e \in \partial_1 v} Z_e$$

Surface Codes with Boundaries

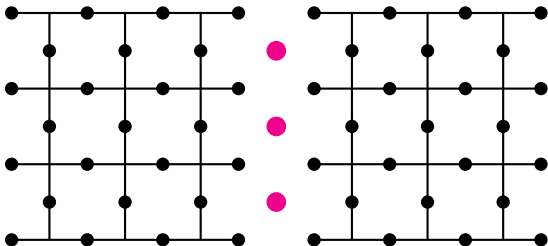


Top and bottom edges are closed boundaries, left and right edges are open boundaries. [3]

Transversal Operations with Planar Surface Codes: Lattice Surgery



Representation of a rough splitting operation on a planar surface code. The initial state is $|0\rangle_L$ and the final state is $|00\rangle_L$.



Truth table for a rough merging operation on a planar surface code.

$\text{In}(1)$	$\text{In}(2)$	Out
0	0	0
0	1	1
1	0	1
1	1	0

Kraus operators for a rough merging operation on a planar surface code:

$$M_R(\rho) = K_{0,R}\rho K_{0,R}^\dagger + K_{1,R}\rho K_{1,R}^\dagger$$

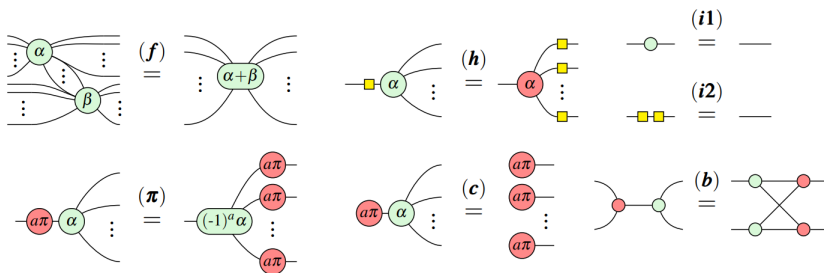
$$K_{0,R} = |+\rangle_M \langle ++| + |-\rangle_M \langle --| \quad K_{1,R} = |+\rangle_M \langle +-| + |-\rangle_M \langle -+|$$

Kraus operator for a rough splitting operation on a planar surface code:

$$U_R = |++\rangle \langle +| + |--\rangle \langle -|$$

ZX Calculus

$$\begin{aligned}
 n \text{ --- } \alpha \text{ --- } m &= \underbrace{|0 \dots 0\rangle}_m \underbrace{\langle 0 \dots 0|}_n + e^{i\alpha} \underbrace{|1 \dots 1\rangle}_m \underbrace{\langle 1 \dots 1|}_n \\
 n \text{ --- } \alpha \text{ --- } m &= \underbrace{|+\dots+\rangle}_m \underbrace{\langle +\dots+|}_n + e^{i\alpha} \underbrace{|-\dots-\rangle}_m \underbrace{\langle -\dots-|}_n
 \end{aligned}$$



Rewrite rules for ZX calculus as presented in [5].

ZX Calculus and Lattice Surgery

Correspondence between ZX diagrams and the Kraus operators of a lattice merging operation [1]:

Diagrammatic representation of the braiding relations for the K_0 and K_1 elements:

- $K_{0,S}$: A green circle on a wire crossing over a red circle on a wire.
- $K_{0,R}$: A red circle on a wire crossing over a green circle on a wire.
- $K_{1,S}$: A red circle on a wire crossing under a green circle on a wire, with a π symbol below the red circle.
- $K_{1,R}$: A green circle on a wire crossing under a red circle on a wire, with a π symbol below the green circle.

Correspondence between ZX diagrams and the Kraus operators for a lattice splitting operation:

 = U_S  = U_R

Logical Verification and Optimization of Quantum Protocols

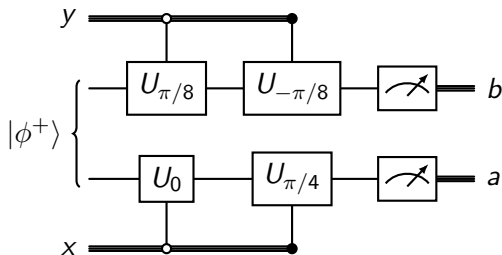
- The rewrite rules of ZX calculus can simplify complex diagrams, conserving the represented linear map and allowing for the logical verification of quantum protocols.
- The simplification of ZX diagrams can also lead to the optimization of quantum protocols, reducing the number of operations and resources required for their implementation.

Quantum Protocols with ZX Calculus

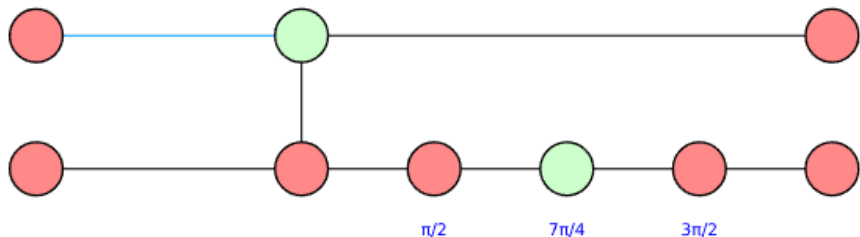
- Quantum Teleportation.
- CHSH Game.
- Magic Square Game.

CHSH Game

CHSH game is a problem in information theory that can be modeled as a non-local cooperative game [8]. Alice and Bob receive each a classical bit of information, x and y , and they must produce a classical bit of information, a and b , respectively, such that they satisfy the winning condition $a \oplus b = x \cdot y$. $P_{\text{classic}} \leq 0.75$.



CHSH Game in ZX Calculus



ZX diagram for the CHSH game with $(x, y) = (0, 0)$ and $(m_0, m_1) = (0, 0)$. The amplitude for each branch:

$$\mathcal{A}_{x,y}(m_0, m_1) = \langle m_0 m_1 | U_{x,y} | 00 \rangle$$

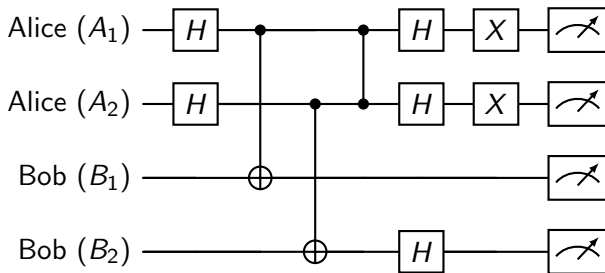
We can verify that $P_{\text{quantum}} \approx 0.85$.

Magic Square Game

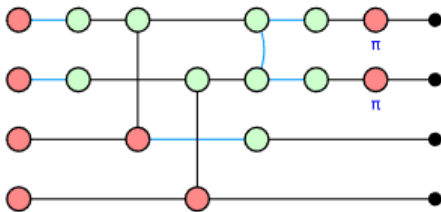
Alice and Bob are given a 3×3 square which they must fill in the following way: Alice is assigned a random row and Bob a random column. They must complete their corresponding cells with values ± 1 , in such a way that the row must contain an odd number of plus signs and the column an odd number of minus signs, and the cell the both share must be filled with the same number by each player. They are not allowed to communicate, but they can share an entangled state. The winning condition is that the bits in the same position in Alice's and Bob's matrices must be equal.

$P_{\text{classic}} \leq 0.875$. Quantum strategy with $P_{\text{quantum}} = 1$ [6]:

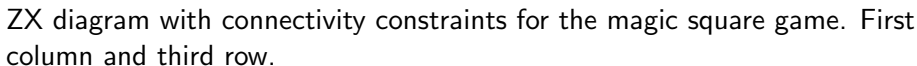
$+I \otimes Z$	$+Z \otimes I$	$+Z \otimes Z$
$+X \otimes I$	$+I \otimes X$	$+X \otimes X$
$-X \otimes Z$	$-Z \otimes X$	$+Y \otimes Y$

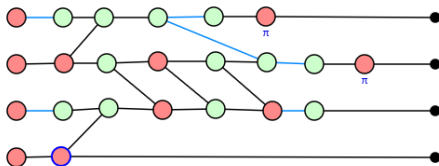


Circuit implementation of the magic square game for the first column and the third row.

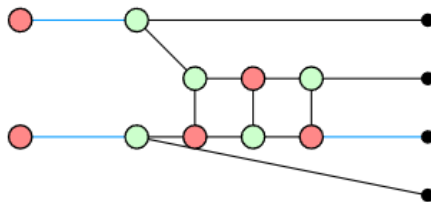


ZX diagram for the magic square game. First column and third row.





ZX diagram with explicit time ordering for the magic square game. First column and third row.



Simplified ZX diagram for the magic square game. First column and third row.

Metric	(0,0)	(0,1)	(0,2)	(1,0)	(1,1)	(1,2)	(2,0)	(2,1)	(2,2)
Original Depth	7	6	8	7	6	8	8	8	8
Original Pathwidth	6	6	6	6	6	6	6	6	7
Optimized Depth	5	6	6	6	6	6	6	6	7
Optimized Pathwidth	5	5	6	5	5	6	6	6	6

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- [1] Niel de Beaudrap and Dominic Horsman. “The ZX calculus is a language for surface code lattice surgery”. In: *Quantum* 4 (2020). arXiv:1704.08670, p. 218.
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References II

- [6] Amine Laghaout et al. “A demonstration of contextuality using quantum computers”. In: *European Journal of Physics* 43.5 (2022), p. 055401. ISSN: 1361-6404. DOI: 10.1088/1361-6404/ac79e0. URL: <http://dx.doi.org/10.1088/1361-6404/ac79e0>.
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